

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology, Kanpur	9.03/10
2021	CBSE(XII)	DAV Public School, Faridabad	95.6%
2019	CBSE(X)	DAV Public School, Faridabad	96.8%

Awards and Achievements

- Received the **Academic Excellence Award** for exceptional academic performance in **2022** at IIT Kanpur
- Bagged a **Silver Medal** in **Inter IIT Tech Meet 11.0** in DevRev's NLP problem as a part of IIT Kanpur's team
- Secured an **All India Rank 2936** in **Joint Entrance Examination(Advanced) 2021** among the 1.4 Lakh candidates
- Secured an **All India Rank 1243** from the 9 lakh applicants in **Joint Entrance Examination(Mains) 2021**

Future Internships

- Incoming Summer **Software Development Intern** at **Atlassian** in 2024

Key Projects

- Expert Answers in a Flash, Inter IIT Tech Meet 11.0** |  (Dec'22-Feb'23)
 - Engineered an **NLP pipeline** to answer a query from a knowledge base in **1 second** on CPU and **20ms** on GPU
 - Leveraged the **Huggingface** library to work with **transformer architecture** models like **MiniLM**, **ELECTRA** etc
 - Implemented a custom **retrieval algorithm** based on **Nearest Neighbour Search** on **Sentence Embeddings**
 - Utilized libraries like **FAISS** and **ANNOY** to build an index and perform **similarity search** on the knowledge base
 - Generated a **synthetic dataset** by extracting pairs of **similar and dissimilar sentences** from given dataset
 - Fine tuned** the MPNet embedding model on **synthetically generated dataset** using **contrastive learning**
 - Quantized** the transformer models using **ONNX Runtime** and **Intel OpenVINO** to reduce inference time
 - Achieved paragraph accuracy of **0.878** and overall F1 score of **0.872** on the evaluation dataset
- IITK Traveller, Programming Club** |  (Dec'22-Feb'23)
 - Designed and developed a unique **esoteric programming language** based on **graphs**, with an unconventional syntax
 - Implemented a custom **interpreter** for the language using **Rust**, leveraging its performance and safety features
 - Wrote a **transpiler** to convert **Super Stack!** esolang programs to IITK Traveller code, **proving turing completeness**
 - Developed a **VSCode Extension** in **typescript** which adds support like auto completion for the language in VSCode
 - Deployed** the interpreter in an **online playground** by developing a frontend using **ReactJS** and a backend using the **Actix** web framework, enabling users to interactively write and execute programs in a user-friendly web environment
 - Containerized** the entire application using **dockerfiles** and **docker compose** for easy deployment on a remote VM
- Control Layer, AUV-IITK** | *Prof. Indranil Saha* |  (Jun'22-Feb'23)
 - Developed and implemented the **control stack** for an Autonomous Underwater Vehicle to enhance its stability
 - Utilized the **Fuzzylite** library in **C++** to develop an **Adaptive PID** algorithm, for stability in dynamic environments
 - Integrated the controller using **ROS**, enabling communication between the control layer and the navigation layer
- CFLockout, Self Project** |  (Jun'23-Present)
 - Developed a lockout style coding **competition platform** based on Codeforces using the **Codeforces API**
 - Used **Next.js** for the frontend, with **Material UI** for responsive components and **Zustand** for central state management
 - Leveraged the **gin gonic** framework to write the backend server in **golang** with **postgres** as the database
 - Implemented a token based **authentication** and **authorization** system using **JSON Web Tokens(JWT)**
 - Utilized **go routines** to achieve **concurrency** for handling multiple sessions, optimizing the platform's performance

Technical Skills

- Programming Languages:** C, C++, HTML, Python, Javascript, CSS, Rust, Go
- Web:** React.js, Express.js, jQuery, Gin, Next.js
- Misc:** Linux, Git, Bash, Docker, ROS, L^AT_EX, Pytorch

Positions of Responsibility

- Technical Team Member, AUV-IITK** (Jun'2022-Present)
 - Selected among a team of 40 undergraduates to participate in international marine robotics competitions like RoboSub
 - Responsible for improving the software stack and design of our second gen AUV-Anahita
- Secretary, Programming Club IITK** (Aug'22-May'23)
 - Among the 25 selected secretaries responsible for managing various activities and sessions pertaining to the club
 - Delivered lectures to 100+ students on fundamentals of programming like OOP and Web Development
 - Mentoring a group of 10 students in learning the basics of NLP through implementing various research papers

Relevant Courses

Data Structures and Algorithms	Signals, Systems and Networks
Fundamentals of Computing	Control System Analysis
Introduction to Electronics	Microelectronics
Probability and Statistics	Linear Algebra and Differential Equations